

Robotics 2

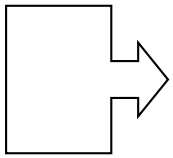
Control in Cartesian Space (or in Task Space)

Prof. Alessandro De Luca

Dipartimento di Ingegneria Informatica
Automatica e Gestionale Antonio Ruberti



Sapienza
Università di Roma



Regulation of robot Cartesian pose



- "PD +" type control for regulation problems
  - proportional to the Cartesian pose error $e_y = y_d - y$, with derivative (velocity) term + cancellation of gravity in joint space
- robot

  dimension of spaces
  - dynamics $M(q)\ddot{q} + c(q,\dot{q}) + g(q) = \tau$ joint = n
  - kinematics $y = k(q) \;\longrightarrow\; \dot{y} = J(q)\dot{q}$ Cartesian = $m \leq n$
- goal: asymptotic stabilization of robot end-effector pose

$$y = y_d, \quad \dot{q} = \dot{q}_d = 0 \;\longrightarrow\; \dot{y}_d = 0$$

if $m = n$, then $\dot{q} = 0 \Leftrightarrow \dot{y} = 0$ up to singularities

if $m < n$, then the goal is not uniquely associated to a complete robot state: $n - m$ joint coordinates are missing …

here $y = y_e = (p_e, \phi_e) = k_e(q)$, with $m \leq 6 \;\leftrightarrow$ end-effector Cartesian space

results equally apply to any $y = k(q)$ and generic $m \;\leftrightarrow$ task space control

A first regulation law

$$(*) \qquad \tau = J^T(q)K_P(y_d - y) - K_D\dot{q} + g(q) \qquad K_P, K_D > 0 \text{ (symmetric)}$$

Theorem

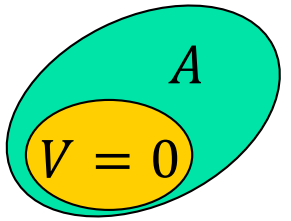
the control law $(*)$ will let the robot state converge asymptotically to the set

$$A = \{\dot{q} = 0, q: K_P\left(y_d - k(q)\right) \in \mathcal{N}(J^T(q))\} \supseteq \{\dot{q} = 0, q: k(q) = y_d\}$$

Proof

from $e_y = y_d - y$ (task/Cartesian error) and an associated Lyapunov-like candidate function

$$V = \frac{1}{2}\dot{q}^T M(q)\dot{q} + \frac{1}{2}e_y^T K_P e_y$$



with $V = 0 \Leftrightarrow (q, \dot{q}) \in \{\dot{q} = 0, q: k(q) = y_d\} \subseteq A$

Proof (cont)

differentiating $V = \frac{1}{2}\dot{q}^T M(q)\dot{q} + \frac{1}{2}e_y^T K_P e_y \geq 0$

$$\dot{V} = \dot{q}^T\left(M\ddot{q} + \frac{1}{2}\dot{M}\dot{q}\right) - e_y^T K_P \dot{y}$$

$\dot{e}_y = -\dot{y}$

$c(q,\dot{q}) = N(q,\dot{q})\dot{q}$ with any factorization

$$= \dot{q}^T\left(\tau - N\dot{q} - g + \frac{1}{2}\dot{M}\dot{q}\right) - e_y^T K_P \dot{y}$$

$$= \dot{q}^T\left(J^T K_P e_y - K_D\dot{q} + g - g\right) - e_y^T K_P J\dot{q}$$

$$= -\dot{q}^T K_D \dot{q} \leq 0$$

with $\dot{V} = 0 \Leftrightarrow \dot{q} = 0$

in these conditions, the closed-loop equations become

$$M(q)\ddot{q} + g(q) = J^T(q)K_P e_y + g(q) \quad \Rightarrow \quad \ddot{q} = M^{-1}(q)J^T(q)K_P e_y$$

$$\Rightarrow \quad \ddot{q} = 0 \Leftrightarrow K_P e_y \in \mathcal{N}(J^T(q))$$

by applying LaSalle theorem, the thesis follows

Corollary

for a given initial state $(q(0), \dot{q}(0))$, if the robot does not encounter any singularity of $J^T(q)$ (configurations where $\rho(J^T) < m \leq n$) during its motion, then there is asymptotic stabilization to one single state (when $m = n$) or to a set of states (when $m < n$) such that

$$e_y = 0, \dot{q} = 0$$

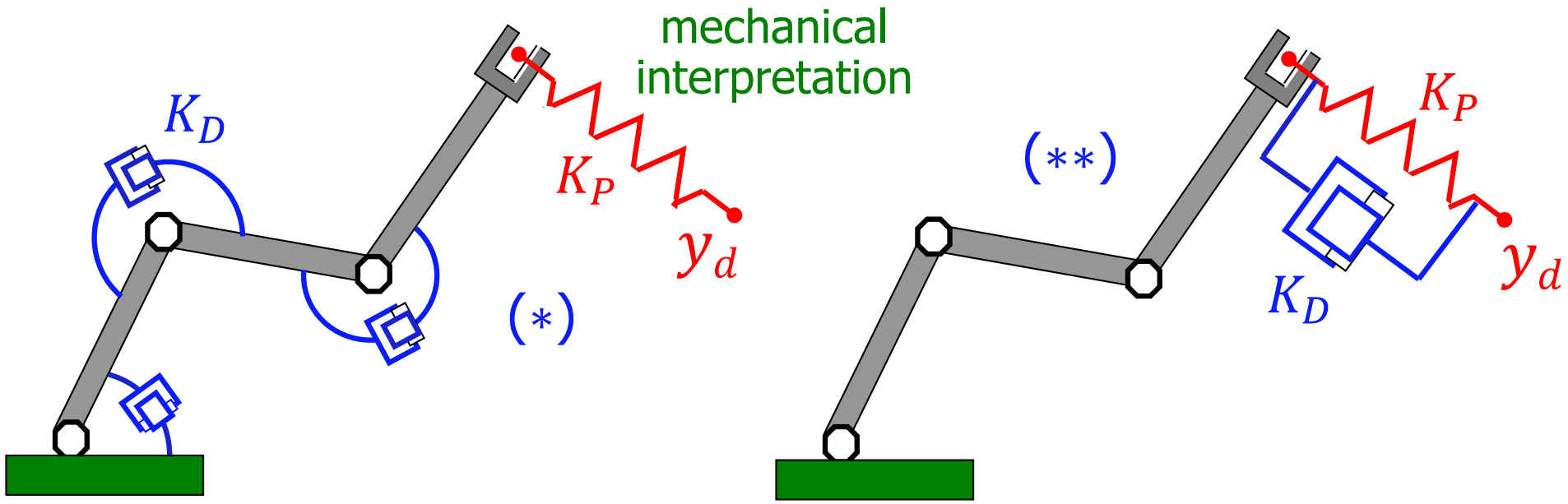
note: singular configurations q of $J^T(q)$ coincide with those of $J(q)$

A possible variant for regulation

"all Cartesian" PD control + gravity cancellation in joint space

$$(**) \quad \tau = J^T(q)(K_P(y_d - y) - K_D\dot{y}) + g(q)$$

$K_P, K_D > 0$ (symmetric)



J^T transforms the "virtual" elastic, for $(*)$, or visco-elastic, for $(**)$, force/torque acting on the end-effector into control torques at the joints

Feedback linearization in task space

robot $M(q)\ddot{q} + c(q,\dot{q}) + g(q) = \tau$

output $y = k(q)$ e.g., end-effector position/orientation

assume $m = n$

algorithm differentiate the output(s) as many times as needed up to the appearance of (at least one of) the input torque(s), then verify if it is possible to solve for the input = "inversion"

(uniform) "relative degree" = 2 for all outputs

$$y = k(q)$$
$$\dot{y} = J(q)\dot{q}$$
$$\ddot{y} = J(q)\ddot{q} + \dot{J}(q)\dot{q}$$
$$= J(q)M^{-1}(q)\big(\tau - c(q,\dot{q}) - g(q)\big) + \dot{J}(q)\dot{q}$$

from dynamic model

Theorem

for a non-redundant robot, it is possible to exactly linearize and decouple the dynamic behavior at the task level if and only if

$$\det J(q) \neq 0$$

Feedback linearization in task space
(in the right coordinates!)



control law $\tau = M(q)J^{-1}(q)a + c(q,\dot{q}) + g(q) - M(q)J^{-1}(q)\dot{J}(q)\dot{q}$

$= \alpha(q,\dot{q}) + \beta(q)a$

$\ddot{y} = J(q)M^{-1}(q)\left(\tau - c(q,\dot{q}) - g(q)\right) + \dot{J}(q)\dot{q} = a$

m chains of double integrators (in task space)

$y, \dot{y}$ are the so-called "linearizing" coordinates

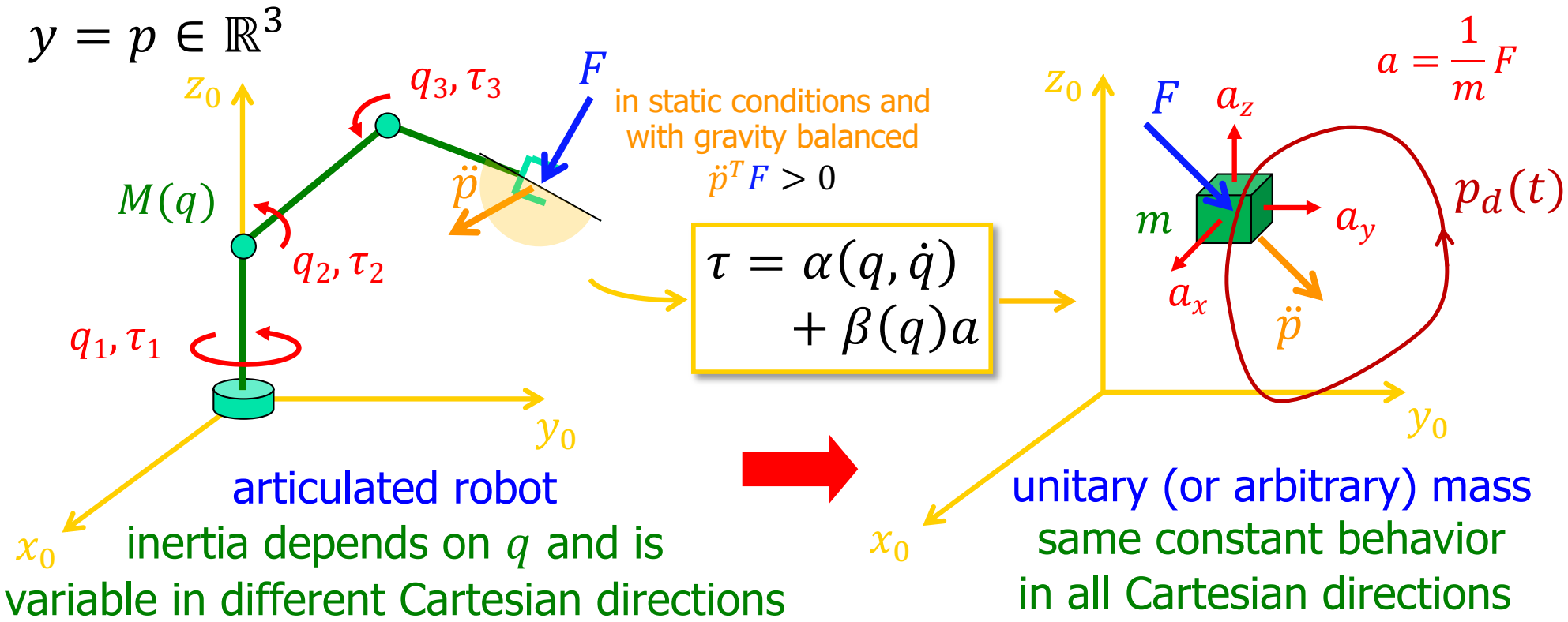
closed-loop equations (in joint space)

$M^{-1} *$ $M\ddot{q} + c + g = M \cdot J^{-1}\left(a - \dot{J}\dot{q}\right) + c + g$

$\ddot{q} = J^{-1}(q)a - J^{-1}(q)\dot{J}(q)\dot{q}$

purely kinematic equations in $q, \dot{q}$
(still nonlinear and coupled!)

Physical interpretation



when a force F is applied at the robot end-effector

- the uncontrolled end-effector will accelerate with $\ddot{p}$ in a different direction

on the other hand, for the robot under Cartesian FBL

- a mass m accelerates always in the same direction of the applied force F

Alternative derivation
in purely task/Cartesian terms



the previous exact linearizing and decoupling law can be rewritten in task/Cartesian terms using a control force/torque (wrench) F

$$\tau = M(q)J^{-1}(q)a + \left(c(q,\dot{q}) - M(q)J^{-1}(q)\dot{J}(q)\dot{q}\right) + g(q)$$

joint torque τ is moved to the task space as $F = J^{-T}(q)\tau$ (for $m = n$)

$$F = (J^{-T}MJ^{-1})a \longrightarrow \text{task inertia} = (JM^{-1}J^{T})^{-1} = M_y$$
$$+ J^{-T}\left(c - MJ^{-1}\dot{J}\dot{q}\right) \rightarrow \text{task Coriolis/centrifugal terms}$$
$$+ J^{-T}g \longrightarrow \text{task gravity}$$
$$= M_y\, a + c_y + g_y$$

this is the feedback linearization law applied to the **task/Cartesian dynamic model** of the robot

$$M_y(q)\ddot{y} + c_y(q,\dot{q}) + g_y(q) = F$$

$$\ddot{y} = a$$

Remarks - 1



- the design of a task trajectory control is completed by stabilizing the tracking error in the m independent chains of double integrators, i.e., by setting

$$a_i = \ddot{y}_{di} + K_{Di}(\dot{y}_{di} - \dot{y}_i) + K_{Pi}(y_{di} - y_i)$$

scalar gains
$K_{Pi} > 0, K_{Di} > 0$
$i = 1, \dots, m$

- the transient behavior of the task error along a desired trajectory is exponentially stable (with arbitrary eigenvalues assigned by choosing the diagonal gains of K_P, K_D)

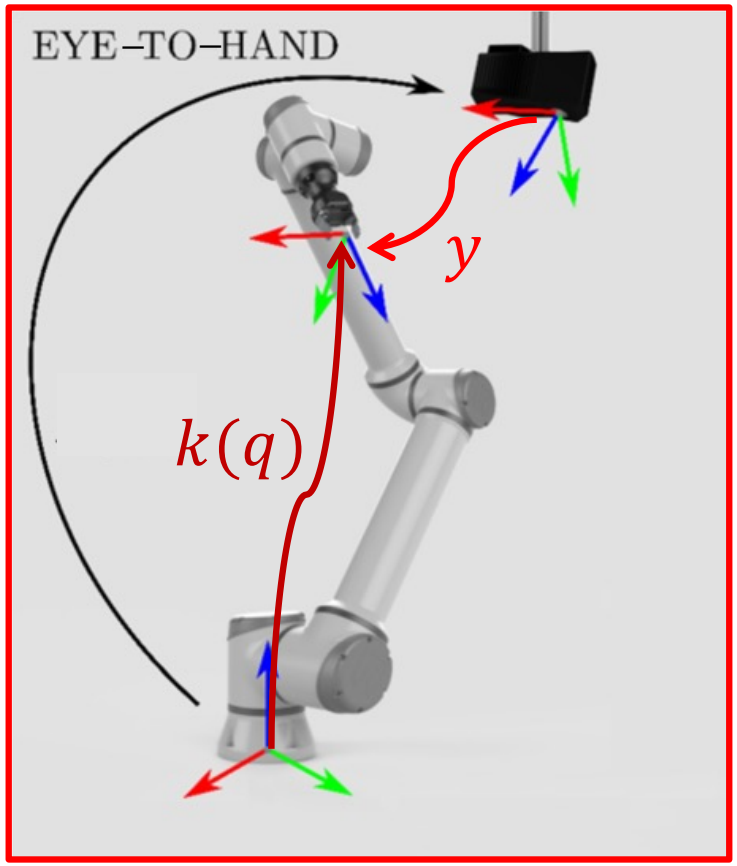
- for y_d = constant (regulation task), the control law becomes

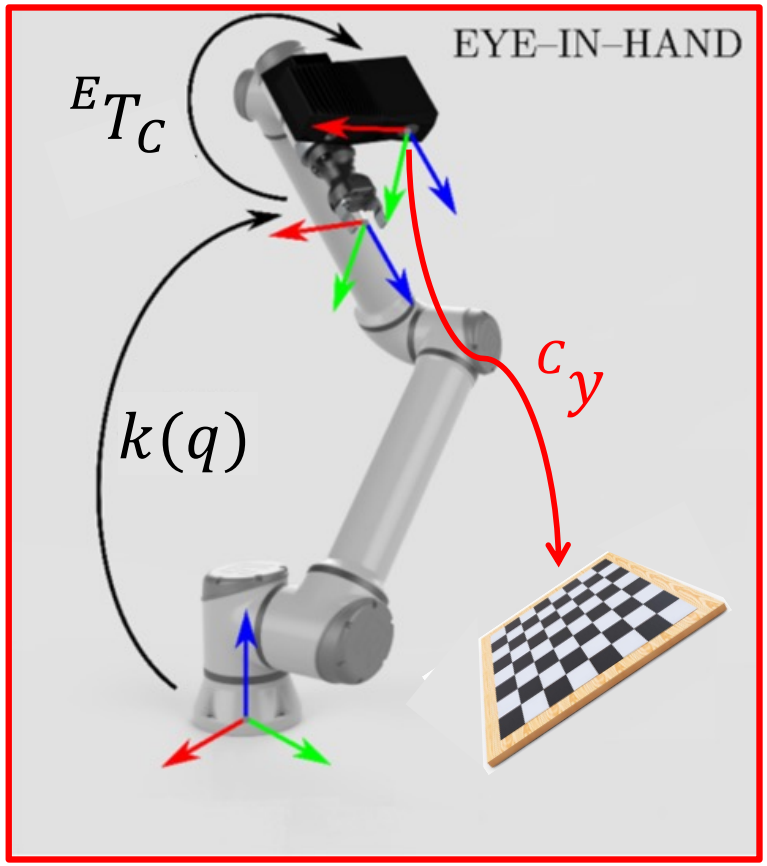
$$\tau = M(q)J^{-1}(q)\left(K_P e_y - K_D J(q)\dot{q}\right) + c(q,\dot{q}) + g(q) - M(q)J^{-1}(q)\dot{J}(q)\dot{q}$$

computationally heavier than a control law designed directly for regulation, such as laws $(*)$ or $(**)$, but retaining the property of an exponentially stable transient error in the task space

Remarks - 2

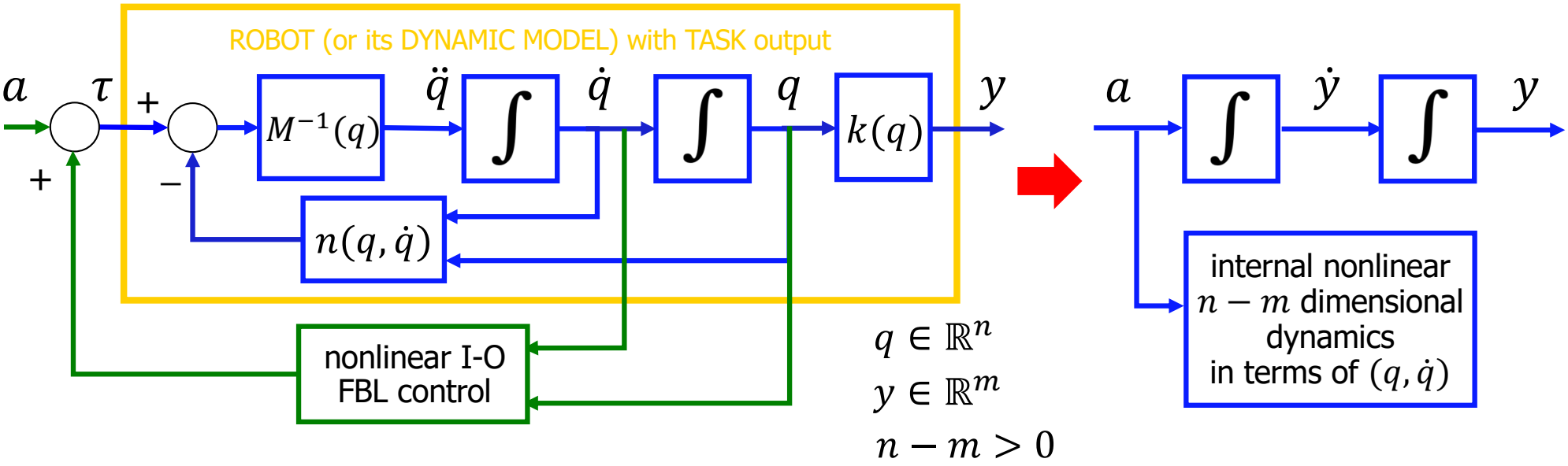
- the Cartesian/task pose (and velocity) is either directly measured by external sensors (e.g., cameras eye-to-hand/eye-in-hand) or computed through the direct (and differential) robot kinematics





Remarks - 3

- in redundant robots ($m < n$), by replacing $MJ^{-1} = (JM^{-1})^{-1}$ in the control law with some (weighted) pseudoinverse $(JM^{-1})^{\#}_{W}$, one still obtains input-output (I-O) decoupling and linearization, but not an exact linearization of the whole state dynamics
  - there is an additional internal dynamics left of dimension $n - m$
  - its stability should be enforced (by some "null-space" torque τ_N)



More on the redundant case …

- suppose $m < n$, but still with a Jacobian J of full rank m
- let the control law (with null-space torque term τ_N) be defined as

$$\tau = (J(q)M^{-1}(q))^{\#}_{W}\left(a - \dot{J}(q)\dot{q} + J(q)M^{-1}(q)\big(c(q,\dot{q}) + g(q)\big)\right) + \left(I - \big(J(q)M^{-1}(q)\big)^{\#}_{W}\, J(q)M^{-1}(q)\right)\tau_N$$

  where $(JM^{-1})^{\#}_{W} = W^{-1}M^{-1}J^T(JM^{-1}W^{-1}M^{-1}J^T)^{-1}$

- three standard choices for $W > 0$

$$W = I \implies (JM^{-1})^{\#} = M^{-1}J^T(JM^{-2}J^T)^{-1}$$
$$W = M^{-1} \implies (JM^{-1})^{\#}_{M^{-1}} = J^T(JM^{-1}J^T)^{-1}$$
$$W = M^{-2} \implies (JM^{-1})^{\#}_{M^{-2}} = M\,J^T(J\,J^T)^{-1} = M\,J^{\#}$$

each associated control torque optimizes a different criterion (see slides on redundant robots!)

- all give the same $\ddot{y} = a$, with τ_N available for null-space control

Conclusions



- most control laws presented in joint space (i.e., driven by a joint error) can be translated with relative ease to the Cartesian space, e.g.
  - regulation with constant gravity compensation
  - adaptive regulation
  - adaptive control for trajectory tracking
- the main issues are related to
  - kinematic singularities, both for the Jacobian transpose and the Jacobian inverse control laws: suitable modifications are needed to obtain singularity robustness
  - kinematic redundancy ($m < n$): use of a stabilizing null-space torque control is needed for the extra $n - m$ generalized coordinates (locally, $n - m$ joint variables)